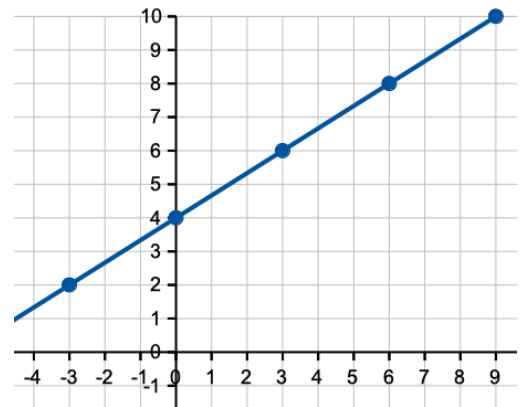


## Creating a Linear Equation from Points

We can figure out the slope from a graph now, by inspecting the rise and run.

We can also **calculate** the slope with a formula, if we're given two points on the line. We call slope "m":

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$



In the graph above, it is clear that the y-intercept is

So the **equation of the line** is:

The generalized form of this equation is  $y = mx + b$

If we can't see the y-intercept ourselves, we can calculate it by plugging in **one point** and **the slope** into  $y = mx + b$  and solving.

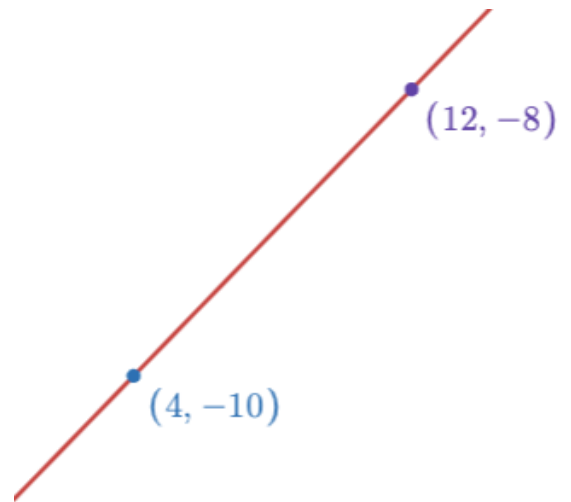
### Practice

- Find the equation for the line that passes through the points (-3,2) and (6,8).

2. For a line that passes through both A(7,22) and B(-4,-11):
- a) Find the slope.
  - b) Solve for  $b$ , the y-intercept.
  - c) State the equation of the line.
3. Find the equation for the line that passes through the points (-2,10) and (8,2).
4. Find the equation for the line that passes through the points (8,0) and (0,-4)

5. Find the equation of a line that passes through C(8,-7) and D(1,7):

6. Find the equation of this line:



7. Find the equation of a line that passes through these two points:

$(3, -1)$

$(12, -7)$

8. A line passes through the points  $(\frac{1}{2}, \frac{1}{3})$  and  $(\frac{1}{4}, -\frac{1}{12})$   
Determine the equation of the line.

9. To host a children's party at an indoor playground, a family pays a non-refundable deposit plus an additional amount for each child attending.

If 10 children attend, the total cost is \$380.

If 15 children attend, the total cost is \$440.

a) Determine the additional cost per child.

b) Determine the initial fee.

c) Write an equation to relate the total cost (C) to the number of children attending (n).

d) What values can be used for n in your equation?